

D791

Integrating AI with Modern Software Applications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A fraud model was trained on 2023 data. By 2026, the share of transactions from mobile wallets has tripled, but fraud patterns for each transaction type look the same. The model runs on a Kubernetes cluster. What kind of change is this?

- A. No drift, because the platform is stable.
- B. Concept drift.
- C. Training-serving skew.
- D. Data drift.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Customer inputs look just as they did at training time, but after a competitor's price cut, the behaviors that signal churn have changed. What kind of drift is this, and how is it detected?

- A. Concept drift, seen as falling accuracy on newly labeled outcomes.
- B. No drift, since the inputs have not changed.
- C. Data drift, seen in input distribution monitors.
- D. Concept drift, seen from input statistics alone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a Python training notebook, income was divided by 1,000 before training. The production Java service passes raw income to the model. Offline accuracy was 92%, but production predictions are wild. The service was written by a separate team in another building. What is the fix?

- A. Retrain the model on more data.
- B. Treat it as overfitting and add regularization.
- C. Use one shared feature transformation for both training and serving.
- D. Move the two teams into the same building.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retailer wants to email personalized product picks to 2 million customers every Monday morning. The model is a deep network trained on GPUs. Which inference approach fits best?

- A. A real-time API called as each email is sent.
- B. Batch scoring ahead of the Monday send.
- C. On-device inference on customers' phones.
- D. Real-time, because deep networks require it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which requirement forces real-time (online) inference rather than batch?

- A. A weekly risk report for managers.
- B. A very large model.
- C. Every modern application always needs real-time inference.
- D. Scoring a card transaction within milliseconds, using details known only at purchase time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A team replaced recommendation model v3 with v4 by overwriting the model file on the server. Click-through fell 15%, and no one can restore v3. The app is built on a popular web framework. What practice prevents this?

- A. Never update the model once it is live.
- B. Back up the server weekly.
- C. Keep models as versioned artifacts in a registry and deploy by version.
- D. Move to a different web framework.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A checkout page must respond within 300 ms end to end. Network time and other services take 180 ms. The fraud model runs on a GPU bought in 2024. What is the most time the model can take?

- A. 300 ms
- B. No limit, because it runs on a GPU.
- C. 480 ms
- D. 120 ms

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A model takes 90 ms, and the rest of the request path takes 160 ms. The target is 200 ms. How many milliseconds must be cut?

- A.** 110 ms
- B.** 450 ms
- C.** 50 ms
- D.** 90 ms

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Each model replica handles 50 requests per second. Peak traffic is 900 requests per second, and replicas must run at no more than 80% of capacity. The model took 6 hours to train. How many replicas are needed?

- A.** 23
- B.** 22
- C.** It depends on the 6-hour training time.
- D.** 18

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A service handles a peak of 1,800 requests per second on 12 replicas, each running at 75% of capacity. What is each replica's full capacity?

- A.** 150 requests per second
- B.** 200 requests per second
- C.** 112.5 requests per second
- D.** 2,400 requests per second

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A support app lets a language model write SQL from customer questions and runs it directly on the production database with an admin account. The model scored well in a vendor benchmark. What is the most important control to add?

- A. Treat the output as untrusted: validate it and run it with read-only access to approved views.
- B. Rely on the strong benchmark score.
- C. Switch to a larger model.
- D. Add a disclaimer to the app’s answers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A voice feature was trained on 5,000 hours of studio-microphone recordings, labeled by a top-rated firm. Customers use it in noisy cars and complain it fails. What is the core data problem, and what helps?

- A. Add more studio recordings.
- B. The data does not represent real use; add recordings from noisy car conditions.
- C. Nothing; 5,000 hours is always enough.
- D. Relabel the data with a different firm.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A fraud model was trained on 2023 data. By 2026, the share of transactions from mobile wallets has tripled, but fraud patterns for each transaction type look the same. ~~The model runs on a Kubernetes cluster.~~ What kind of change is this?

Target word: classify the drift **Decides:** The input mix changed; the input-outcome relationship did not.

A	Camouflage	The hosting platform does not decide drift.
B	Confusion set	The meaning of the inputs for fraud has not changed.
C	Adjacent category	Skew is a processing mismatch, not a real-world shift.
D	Correct	The distribution of inputs shifted.

2. Answer A Concept 1 • transfer

Customer inputs look just as they did at training time, but after a competitor's price cut, the behaviors that signal churn have changed. What kind of drift is this, and how is it detected?

Target word: name the drift and its signal **Decides:** The relationship changed; inputs did not.

A	Correct	Only outcome-based metrics reveal a changed relationship.
B	Dropped element	Ignores the changed relationship to the outcome.
C	Inverted operation	Inputs are stable, so input monitors stay quiet.
D	Confusion set	Input statistics cannot show a changed meaning.

3. Answer C Concept 2

In a Python training notebook, income was divided by 1,000 before training. The production Java service passes raw income to the model. Offline accuracy was 92%, but production predictions are wild. ~~The service was written by a separate team in another building.~~ What is the fix?

Target word: diagnose and fix **Decides:** Training and serving prepare the feature differently.

A	Adjacent category	The model is fine; its production inputs are wrong.
B	Confusion set	Offline results were good; the gap is in data preparation.
C	Correct	Removes training-serving skew at its source.
D	Camouflage	Location does not make the code consistent.

4. Answer B Concept 3

A retailer wants to email personalized product picks to 2 million customers every Monday morning. ~~The model is a deep network trained on CPUs.~~ Which inference approach fits best?

Target word: choose inference mode **Decides:** Predictions can be made before they are needed.

A	Inverted operation	Adds latency and cost with no benefit.
B	Correct	Known schedule and no request-time data make batch efficient.
C	Adjacent category	Solves privacy or offline needs, not a scheduled email.
D	Camouflage	Model type does not decide timing.

5. Answer D Concept 3 · transfer

Which requirement forces real-time (online) inference rather than batch?

Target word: trigger for real-time **Decides:** Needs request-time data within a short deadline.

A	Inverted operation	A classic batch use case.
B	Confusion set	Size affects cost, not whether predictions can be precomputed.
C	Absolute language	Many uses are cheaper and simpler in batch.
D	Correct	The input does not exist until the request arrives.

6. Answer C Concept 4

A team replaced recommendation model v3 with v4 by overwriting the model file on the server. Click-through fell 15%, and no one can restore v3. ~~The app is built on a popular web framework.~~ What practice prevents this?

Target word: enable safe model updates **Decides:** No versioned copy of the old model.

A	Absolute language	Models need updates as data changes.
B	Adjacent category	Backups are coarse and not tied to model versions or deploys.
C	Correct	Rollback becomes a one-step switch to v3.
D	Camouflage	The framework is not the cause.

7. Answer D Concept 5

A checkout page must respond within 300 ms end to end. Network time and other services take 180 ms. ~~The fraud model runs on a GPU bought in 2024.~~ What is the most time the model can take?

Target word: model latency budget **Decides:** 300 – 180.

A	Wrong stem number	The whole target, leaving nothing for the other steps.
B	Camouflage	Hardware does not change the budget.
C	Inverted operation	Adds instead of subtracting.
D	Correct	$300 - 180 = 120$ ms.

8. Answer C Concept 5 · transfer

A model takes 90 ms, and the rest of the request path takes 160 ms. The target is 200 ms. How many milliseconds must be cut?

Target word: solve for the overrun **Decides:** Total 250 ms versus 200 ms.

A	Confusion set	$200 - 90$ is the budget for other steps, not the overrun.
B	Inverted operation	Adds all three numbers.
C	Correct	$(90 + 160) - 200 = 50$ ms.
D	Wrong stem number	The model's whole time.

9. Answer A Concept 6

Each model replica handles 50 requests per second. Peak traffic is 900 requests per second, and replicas must run at no more than 80% of capacity. ~~The model took 6 hours to train.~~ How many replicas are needed?

Target word: replica count **Decides:** $900 \div (50 \times 0.8)$, rounded up.

A	Correct	$900 / 40 = 22.5$, rounded up to 23.
B	Inverted operation	Rounds down, leaving peak traffic under-served.
C	Camouflage	Training time does not affect serving capacity.
D	Dropped element	$900 / 50$ drops the 80% cap.

10. Answer B Concept 6 · transfer

A service handles a peak of 1,800 requests per second on 12 replicas, each running at 75% of capacity. What is each replica's full capacity?

Target word: solve for per-replica capacity **Decides:** Load per replica $\div 0.75$.

A	Dropped element	Drops the 75% utilization.
B	Correct	$1,800 / 12 = 150$; $150 / 0.75 = 200$.
C	Inverted operation	Multiplies by 0.75 instead of dividing.
D	Confusion set	Total fleet capacity, not per replica.

11. Answer A Concept 7

A support app lets a language model write SQL from customer questions and runs it directly on the production database with an admin account. ~~The model scored well in a vendor benchmark.~~ What is the most important control to add?

Target word: key risk control **Decides:** Model output executes with full privileges.

A	Correct	Constraining output and privileges limits the damage from any bad query.
B	Camouflage	Benchmarks do not remove the need for controls.
C	Adjacent category	A better model still produces some bad output.
D	Dropped element	A notice does not stop harmful queries from running.

12. Answer B Concept 8

A voice feature was trained on 5,000 hours of studio-microphone recordings, ~~labeled by a top-rated firm.~~ Customers use it in noisy cars and complain it fails. What is the core data problem, and what helps?

Target word: diagnose the data gap **Decides:** Training conditions differ from real use.

A	Adjacent category	More of the same unrepresentative data.
B	Correct	Training must reflect production conditions.
C	Absolute language	Quantity does not fix a mismatch in conditions.
D	Camouflage	Labels are not the problem.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Data drift vs. concept drift	Data drift is a change in the inputs' distribution; concept drift is a change in how inputs relate to the outcome, visible only through outcome-based metrics.	Assuming stable inputs mean a healthy model.	What changed: the mix of inputs, or what they mean for the outcome.
2	Training-serving skew	Features must be computed the same way in training and production; one shared transformation removes the gap.	Blaming the model when the serving code prepares data differently.	Great offline results and strange production predictions.
3	Batch vs. real-time inference	Use batch scoring when predictions can be made ahead of need; use real-time inference when the answer depends on data available only at request time.	Choosing real-time because the model is modern or large.	When the prediction is needed and what data it depends on.
4	Model versioning and rollback	Store models as versioned artifacts in a registry and deploy by version so a bad release can be rolled back in one step.	Overwriting the model file in place.	A worse new model and no way to restore the old one.
5	Latency budgets	The model's allowed time = end-to-end target – time used by everything else on the request path.	Giving the model the whole target, or adding instead of subtracting.	A response-time target and the time other steps take.
6	Sizing inference capacity	$\text{Replicas} = \text{peak load} \div (\text{per-replica capacity} \times \text{target utilization})$, rounded up.	Ignoring headroom or rounding down.	Peak requests, per-replica capacity, and a utilization cap.
7	Model output is untrusted input	Validate and constrain AI output before it touches other systems, and give AI features least-privilege access.	Letting a model act with full privileges because it tested well.	Model output flowing straight into a database, command, or action.
8	Representative training data	Training data must reflect the conditions and users the system will meet in production.	Adding more of the same unrepresentative data.	Good lab results, poor results in a different real-world setting.

Course scope (public description): Integrating AI with Modern Software Applications prepares students to analyze and explain various technical and analytical components of machine learning models. Students will explore the design of system architectures to scale, while looking at potential risks and mitigation strategies for integrating AI into existing software applications. The content in this course will give the student the ability to understand how to describe data used by AI systems, its sources, and characteristics, as well as instructing on methods for deployment and monitoring.